One Table, Two Study Designs

Unit 3 · Topic 3.14 · THE PROBLEM

Suppose a transit authority obtains the shown counts under either design.

Design A: Take independent SRSs of 100 from each of three route populations of 5,000 and record ticket method.

Design B: Take one SRS of 300 from 15,000 system riders and record both route and ticket method.

Ari: "Route rows imply homogeneity; the counts are enough to choose."

Bea: "Read how riders were sampled; identical counts can support different population claims."

Ticket method

Route	Mobile	Card	Cash	Total
North	60	25	15	100
East	48	32	20	100
West	42	33	25	100
Total	150	90	60	300

FIRST TAKE — one sentence before you work the parts

Do the same counts mean that riders were selected in the same way? Explain in one sentence.

DOK 1 **(a)** How many random samples are taken in Design A and in Design B?

DOK 2 **(b)** Calculate the three expected counts for the North row. Explain briefly why the other two rows have the same expected counts.

DOK 3 **(c)** In 4 sentences, judge Ari's rule. Name and justify the test for each design, state each null hypothesis in context, and explain why the same table can lead to different test names.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.